

SARRA BOUSNINA

AI Software Engineer

sarra.bousnina@esprit.tn • +216 26 529 858 • [linkedin.com/in/sarra-bousnina](https://www.linkedin.com/in/sarra-bousnina) • github.com/sarrabousnina • sarrabousnina.github.io

Tunis, Tunisia

Professional Summary

Software Engineering student who independently built advanced AI expertise in agentic systems, RAG, multimodal models, and LLMs, validated by 5 AI hackathon wins. I design and ship full-stack AI solutions from concept to cloud. Targeting Master's programs in AI and AI Engineering roles where I can ship scalable, intelligent systems that bridge research and real-world impact.

Experience

AI Engineer Intern End-of-Studies Project, Otraverse, Tunis

Feb 2026 – Present

- Built a full-stack AI pipeline for automated 2D-to-3D scene generation, integrating computer vision models with generative 3D synthesis
- Deployed production-ready FastAPI services on GCP using Docker and Terraform, optimizing inference workflows for performance, scalability, and maintainability

AI Software Development Intern, Mahd.Group, Tunis

Jul – Aug 2025

- Built CorrectMeAI full-stack platform (React.js + Flask) that automates exam grading using OCR and LLMs, enhanced with a custom RAG document retriever and ReAct-style agent for contextual reasoning

Software Development Intern, CTAMA Insurance, Tunis

Jul – Aug 2024

- Built MyCTAMA insurance app (.NET MAUI) claims processing, quote generation, agency locator

Projects & Awards

EagleScout

2nd Prize, CyberIA (33 teams)

AI-powered vulnerability detection using VLM and ReAct-style graph traversal reasoning

BFF Loan Hub

2nd Prize, Finnovo 1.0

AI loan application system with Groq Llama-4-Scout for verification and fraud detection (built and demoed end-to-end in 24 hours)

TimeForge

1st Prize, Bal des Projets 2025

Award-winning productivity app with focus tracking, mood analysis (DeepFace), and AI insights (1st Prize across all engineering specializations at ESPRIT)

AI MINDS

Data Pipeline

Data ingestion pipeline for MemoryOS multimodal assistant (FastAPI, BGE-M3, CLIP, FAISS, Whisper)

InspireAI

Personal SaaS Project

AI content studio with multi-modal generation and ReAct-style AI agent

Additional Awards: 1st Prize – INSAT Hackathon • 2nd Prize – EY Œ Dauphine

Education

Software Engineering (Final Year), ESPRIT, Tunis

2023 – Present

Software Specialization GPA: 15.7/20 Rank 3/class

Pre-Engineering Program (MP), IPEIN, Nabeul

2021 – 2023

Mathematics Baccalaureate with honors, Pioneer High School of Hammam Lif

2017 – 2021

GPA: 15.31/20

Technical Skills

AI: AI Engineering, Agentic AI, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), OCR, Prompt Engineering, Transformers, NLP, 3D Vision, Computer Vision

Languages: Python, Java, JavaScript, TypeScript, C#, SQL

Frameworks: FastAPI, Flask, LangChain, PyTorch, Spring Boot, Angular, React, .NET MAUI

Cloud/DevOps: GCP, Vertex AI, Docker, Terraform, Git, CI/CD, REST APIs

Databases: MySQL, MongoDB, PostgreSQL, FAISS, Qdrant, Vertex AI Vector Search

Certifications

NVIDIA DLI (9): RAG Agents, LLM Apps, Prompt Engineering, Transformer NLP, Anomaly Detection, Multimodal Agents, LLM Evaluation, Deep Learning, Rapid App Dev

Other: AWS Cloud Foundations, Scrum, Supervised ML (Coursera), Hashgraph Developer

Community & Leadership

Mentor @ DeepFlow AI Club • IEEE Student Branch • Volunteer: HackFlow, Engineering Road, Integration Day

Languages

French (Fluent) | Arabic (Native) | English (Fluent) | Spanish (Basic)